

Getting Past the Hype Around Hadoop

The term Big Data and the name Hadoop are bandied about freely in computer circles. In this article, the author attempts to explain them in very simple terms.



Imagine this scenario: You have 1GB of data that you need to process. The data is stored in a relational database in your desktop computer which has no problem managing the load. Your company soon starts growing very rapidly, and the data generated grows to 10GB, and then 100GB. You start to reach the limits of what your current desktop computer can handle. So what do you do? You scale up by investing in a larger computer, and you are then alright for a few more months. When your data grows from 1TB to 10TB, and then to 100TB, you are again quickly approaching the limits of that computer. Besides, you are now asked to feed your application with unstructured data coming from sources like Facebook, Twitter, RFID readers, sensors, and so on. Your managers want to derive information from both the relational data and the unstructured data, and they want this information as soon as possible. What should you do?

Hadoop may be the answer. Hadoop is an open source project of the Apache Foundation. It is a framework written

in Java, originally developed by Doug Cutting, who named it after his son's toy elephant!

Hadoop uses Google's MapReduce technology as its foundation. It is optimised to handle massive quantities of data which could be structured, unstructured or semi-structured, using commodity hardware, i.e., relatively inexpensive computers. This massive parallel processing is done with great efficiency. However, handling massive amounts of data is a batch operation, so the response time is not immediate. Importantly, Hadoop replicates its data across different computers, so that if one goes down, the data is processed on one of the replicated computers.

Big Data

Hadoop is used for Big Data. Now what exactly is Big Data? With all the devices available today to collect data, such as RFID readers, microphones, cameras, sensors, and so on, we are seeing an explosion of data being collected worldwide.